

Uncertainty

Suppose A, B, C, D are **binary** random variables. How many entries/parameters are **needed** to store the tables? Be **efficient** in storage.

(i) $P(A \mid C = \text{true})$

$P(A \mid C = \text{true})$ has 2 entries that sum to 1; one for $P(A = \text{true} \mid C = \text{true})$ and one for $P(A = \text{false} \mid C = \text{true})$.

It is **enough** to store 1 entry $P(A = \text{true} \mid C = \text{true})$, and you can compute $P(A = \text{false} \mid C = \text{true}) = 1 - P(A = \text{true} \mid C = \text{true})$

So, total needed is 1 entry.

(ii) $P(A, D \mid B = \text{true}, C = \text{true})$

The givens B, C are fixed, and we have $2 \times 2 = 4$ settings for A, D that sum to 1.

It is **enough** to store 3 entries and the last entry can be computed as $1 - \text{sum of 3 entries}$.

So, total needed is 3 entries.

(iv) $P(B \mid A = \text{true}, C, D)$

2 entries for $P(B \mid A = \text{true}, C = \text{true}, D = \text{true})$ that sums to 1. It is **enough** to store 1 entry.

2 entries for $P(B \mid A = \text{true}, C = \text{true}, D = \text{false})$ that sums to . It is **enough** to store 1 entry.

2 entries for $P(B \mid A = \text{true}, C = \text{false}, D = \text{true})$ that sums to . It is **enough** to store 1 entry.

2 entries for $P(B \mid A = \text{true}, C = \text{false}, D = \text{false})$ that sums to . It is **enough** to store 1 entry.

So, total needed is $1 + 1 + 1 + 1 = 4$ entries.

(iii) $P(B) \cdot P(A \mid B) \cdot P(C, D)$

1 entry for $P(B)$. 1 entry for $P(A \mid B = \text{true})$. 1 entry for $P(A \mid B = \text{false})$. 3 entries for $P(C, D)$.

So, Total needed is $1 + 1 + 1 + 3 = 6$

(iv) Suppose that we said that A is independent of all other variables, and that C is independent of all other variables given B and D. The remaining variables have no other assumptions. What is the probability distribution formula and how many entries do we need to store?

$$P(A) \cdot P(C | B, D) \cdot P(B, D)$$

1 entry for $P(A)$.

4 entries for $P(C | B, D)$.

3 entries for $P(B, D)$.

So, Total needed is $1 + 4 + 3 = 8$

(v) Consider same assumptions as in (iv), but now consider that B has 5 possible values, and C has 3 possible values. What is the probability distribution formula and how many entries do we need to store?

$$P(A) \cdot P(C | B, D) \cdot P(B, D)$$

1 entry for $P(A)$.

For $P(C | B, D)$, since (B, D) has $5 \cdot 2$ possible settings, and C has 3, then for each setting of (B, D) we need to store 2 entries; so total of $2 \cdot (5 \cdot 2) = 20$ entries.

9 entries for $P(B, D)$.

So, Total needed is $1 + 20 + 9 = 30$

Consider the following joint-probability table

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

Calculate the following queries.

a) $P(\text{toothache})$.

b) $P(\neg \text{cavity})$

c) $P(\text{toothache} \mid \text{cavity})$

0.6

d) $P(\neg \text{cavity} \mid \text{toothache} \vee \text{catch})$

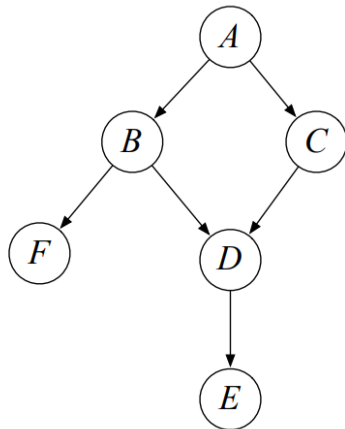
$$= \frac{P((\neg \text{cavity} \wedge \text{toothache}) \vee (\neg \text{cavity} \wedge \text{catch}))}{P(\text{toothache} \vee \text{catch})}$$

$$\begin{aligned} \text{Numerator} &= P(\neg \text{cavity} \wedge \text{toothache}) + P(\neg \text{cavity} \wedge \text{catch}) - P(\neg \text{cavity} \wedge \text{toothache} \wedge \text{catch}) \\ &= 0.08 + 0.16 - 0.016 = 0.224 \end{aligned}$$

$$\begin{aligned} \text{Denominator} &= P(\text{toothache}) + P(\text{catch}) - P(\text{toothache} \wedge \text{catch}) \\ &= 0.416 \end{aligned}$$

$$\text{Numerator/Denominator} = 0.5384$$

In the following Bayes net, which of the following statements are guaranteed true.



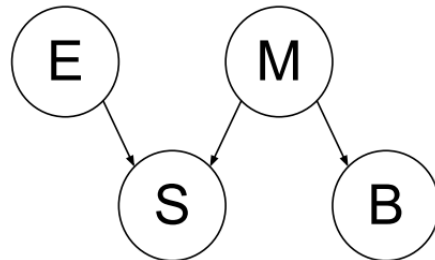
- a) $A \perp\!\!\!\perp E$
- b) $B \perp\!\!\!\perp C \mid A$
- c) $F \perp\!\!\!\perp C \mid A$
- d) $B \perp\!\!\!\perp C \mid A, E$

Use d-separation technique.
b and c are guaranteed true.

Consider the following Bayes' Net.

$P(E)$	
$+e$	0.4
$-e$	0.6

$P(S E, M)$			
$+e$	$+m$	$+s$	1.0
$+e$	$+m$	$-s$	0.0
$+e$	$-m$	$+s$	0.8
$+e$	$-m$	$-s$	0.2
$-e$	$+m$	$+s$	0.3
$-e$	$+m$	$-s$	0.7
$-e$	$-m$	$+s$	0.1
$-e$	$-m$	$-s$	0.9



$P(M)$	
$+m$	0.1
$-m$	0.9

$P(B M)$		
$+m$	$+b$	1.0
$+m$	$-b$	0.0
$-m$	$+b$	0.1
$-m$	$-b$	0.9

- What is $P(-e, -s, -m, -b)$? 0.4374
- What is $P(+b)$? 0.19
- What is $P(+m | +b)$? 0.5263
- What is $P(+m | +s, +b, +e)$? 0.5814
- For this given BN, how many entries are **needed** for storage? Be **efficient**.

$P(E)$ needs 1.

$P(M)$ needs 1.

$P(S | E, M)$ needs 1 for every setting of E and M. So, total is $1*2*2 = 4$

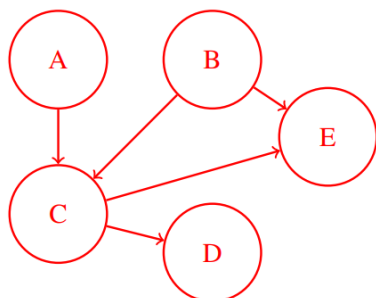
$P(B | M)$ needs 1 for every setting of M. So, total is $1*2 = 2$

So, total entries needed for BN storage is $1 + 1 + 4 + 2 = 8$ entries

Consider a BN defined by

$$P(A) \cdot P(B) \cdot P(C|A, B) \cdot P(D|C) \cdot P(E|B, C)$$

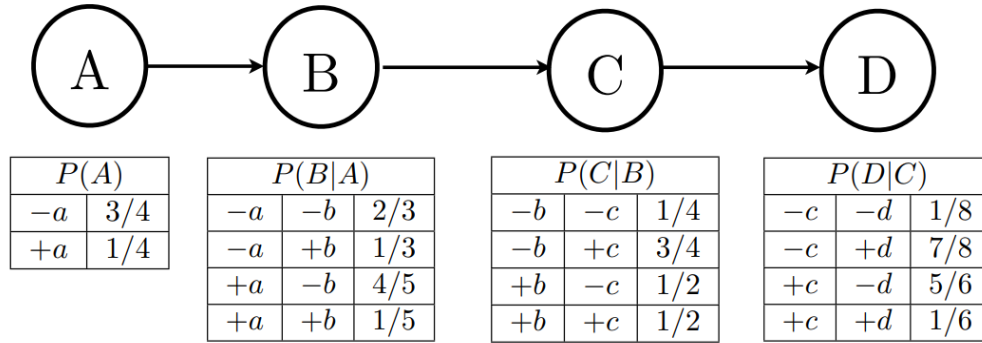
a) Draw the BN associated with this probability distribution:



b) Suppose that the variables in this BN are binary except for C which can take on 5 values. How many entries are needed for storage? Be efficient.

Total entries needed for BN storage is $1 + 1 + 16 + 5 + 10 = 33$ entries

You are given the following BN:



a) You are given the following samples:

$(+a, +b, -c, -d)$ $(+a, -b, +c, -d)$ $(-a, +b, +c, -d)$ $(-a, -b, +c, -d)$	$(+a, -b, -c, +d)$ $(+a, +b, +c, -d)$ $(-a, +b, -c, +d)$ $(-a, -b, +c, -d)$
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(i) If these samples came from doing Prior Sampling, approximate $P(+c)$. 5/8

(ii) Clearly cross out the samples that would not be used when doing Rejection Sampling for this task, and approximate $P(+c \mid +a, -d)$.

$(+a, +b, -c, -d)$ $(+a, -b, +c, -d)$ $(-a, +b, +c, -d)$ $(-a, -b, +c, -d)$	$(+a, -b, -c, +d)$ $(+a, +b, +c, -d)$ $(-a, +b, -c, +d)$ $(-a, -b, +c, -d)$
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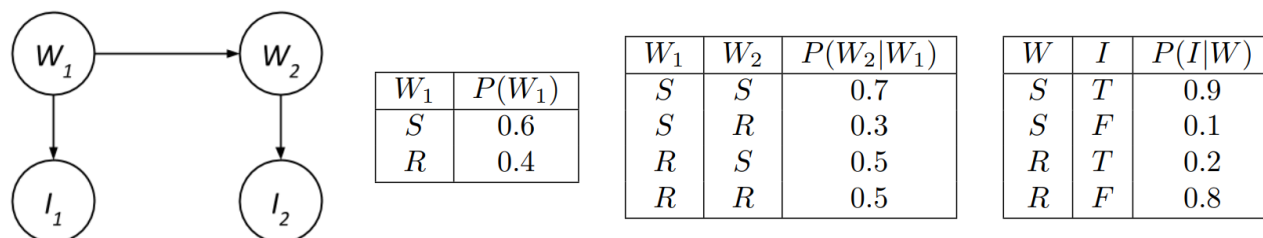
2/3

b) Using Likelihood Weighting Sampling to estimate $P(-a \mid +b, -d)$, the following samples were obtained. What is the weight of each sample?

Sample	Weight
$-a \quad +b \quad +c \quad -d$	$\frac{P(+b \mid -a)P(-d \mid +c)}{P(+b \mid -a)P(-d \mid +c)} = \frac{1/3 * 5/6}{1/3 * 5/6} = 5/18 = 0.277$
$+a \quad +b \quad +c \quad -d$	$\frac{P(+b \mid +a)P(-d \mid +c)}{P(+b \mid +a)P(-d \mid +c)} = \frac{1/5 * 5/6}{1/5 * 5/6} = 5/30 = 1/6 = 0.17$
$+a \quad +b \quad -c \quad -d$	$\frac{P(+b \mid +a)P(-d \mid -c)}{P(+b \mid +a)P(-d \mid -c)} = \frac{1/5 * 1/8}{1/5 * 1/8} = 1/40 = 0.025$
$-a \quad +b \quad -c \quad -d$	$\frac{P(+b \mid -a)P(-d \mid -c)}{P(+b \mid -a)P(-d \mid -c)} = \frac{1/3 * 1/8}{1/3 * 1/8} = 1/24 = 0.042$

c) From the weighted samples, estimate $P(-a \mid +b, -d)$. 0.625

Suppose you are given the following Bayes' Net:



Suppose we produce the following samples of (W_1, I_1, W_2, I_2) from the ice-cream model:

R, F, R, F R, F, R, F S, F, S, T S, T, S, T S, T, R, F
 R, F, R, T S, T, S, T S, T, S, T S, T, R, F R, F, S, T

a) Using the samples, what is approximate $P(W_2 = R)$? 0.5

b) Cross off samples above which are rejected by rejection sampling if we're computing $P(W_2 | I_1 = T, I_2 = F)$

~~R, F, R, F~~ ~~R, F, R, F~~ ~~S, F, S, T~~ ~~S, T, S, T~~ S, T, R, F
~~R, F, R, T~~ ~~S, T, S, T~~ ~~S, T, S, T~~ S, T, R, F ~~R, F, S, T~~

c) Rejection sampling seems to be wasting a lot of effort, so we decide to switch to likelihood weighting.

Assume we generate the following six samples given the evidence $I_1 = T$ and $I_2 = F$:

$(W_1, I_1, W_2, I_2) = \{ (S, T, R, F), (R, T, R, F), (S, T, R, F), (S, T, S, F), (S, T, S, F), (R, T, S, F) \}$

What is the weight of the first sample (S, T, R, F) above? 0.72

d) Use likelihood weighting to estimate $P(W_2 = S | I_1 = T, I_2 = F)$. 0.111

Intro to ML:

Question 1:

Consider the following gathered dataset from a manufacturing process where we want to predict if a material is safe given its Volume, Density, and Shape. The “Is Safe?” column indicates it is safe (1) or **not** safe (0).

Your supervisor has already split the data into training and validation sets.

Training Data

Volume	Density	Shape	Is Safe?
53	93	Sphere	1
62	49	Cylinder	0
49	52	Sphere	0
100	83	Sphere	0
116	92	Sphere	1
4	2	Sphere	0
27	48	Cube	0
51	65	Sphere	1
87	17	Cylinder	1
16	10	Sphere	1

Validation Data

Volume	Density	Shape	Is Safe?
100	95	Sphere	1
141	59	Cylinder	1
0	158	Cylinder	0
92	44	Cylinder	0
43	150	Sphere	1

1.1 What kind of supervised learning task is this?

1.2 Which columns are the **input features** of this supervised learning task?

1.3 Which columns are the **output labels** of this supervised learning task?

1.4 Which of the **input features** are **most likely continuous**?

1.5 Which of the **input features** are **most likely discrete (finite set of values)**?

1.6 **Based on the training set**, how would you use one-hot-encoding on Shape; show the value to code mapping. Apply this to the Validation data's Shape column.

1.7 Now suppose that Shape can only have four possible values {Sphere, Cube, Cylinder, Cone}. How would you use one-hot-encoding on Shape; show the value to code mapping. Apply this to the Validation data's Shape column.

Question 2:

- 3.1 What is the difference between empirical loss and generalization loss?
- 3.2 What is the main goal in machine learning in terms of empirical and generalization loss?
- 3.3 Denote the training set accuracy as Acc_{train} and test set accuracy as Acc_{test} . Explain what **overfitting** and **underfitting** is in terms of these two symbols.
- 3.4 How is dataset splitting done for regression and classification tasks.
- 3.5 Briefly explain the main difference between supervised and unsupervised learning.
- 3.6 Briefly explain the three common tasks in unsupervised learning.

Model Selection and Evaluation

1.1 Briefly describe **generalization** in the context of supervised learning.

1.2 Describe the train-val-test scheme.

1.3 Describe the k-fold CV scheme.

1.4 When is it preferred to use Precision-Recall AUC?

1.5 Describe in words what ROC-AUC measures for a given model.

1.6 A model was trained on a training set for a binary classification task. The model's label predictions on the test set is given in the table below. Fill out the confusion matrix for this model.

feature 1	feature2	True label	model prediction
0.11	0.21	1	1
0.70	0.05	1	0
0.27	0.23	0	0
0.63	0.78	0	1
0.90	0.36	0	0
0.90	0.95	0	1
0.25	0.22	1	0

		Actual class	
		Positive	Negative
Predicted class	Positive	TP:	FP:
	Negative	FN:	TN:

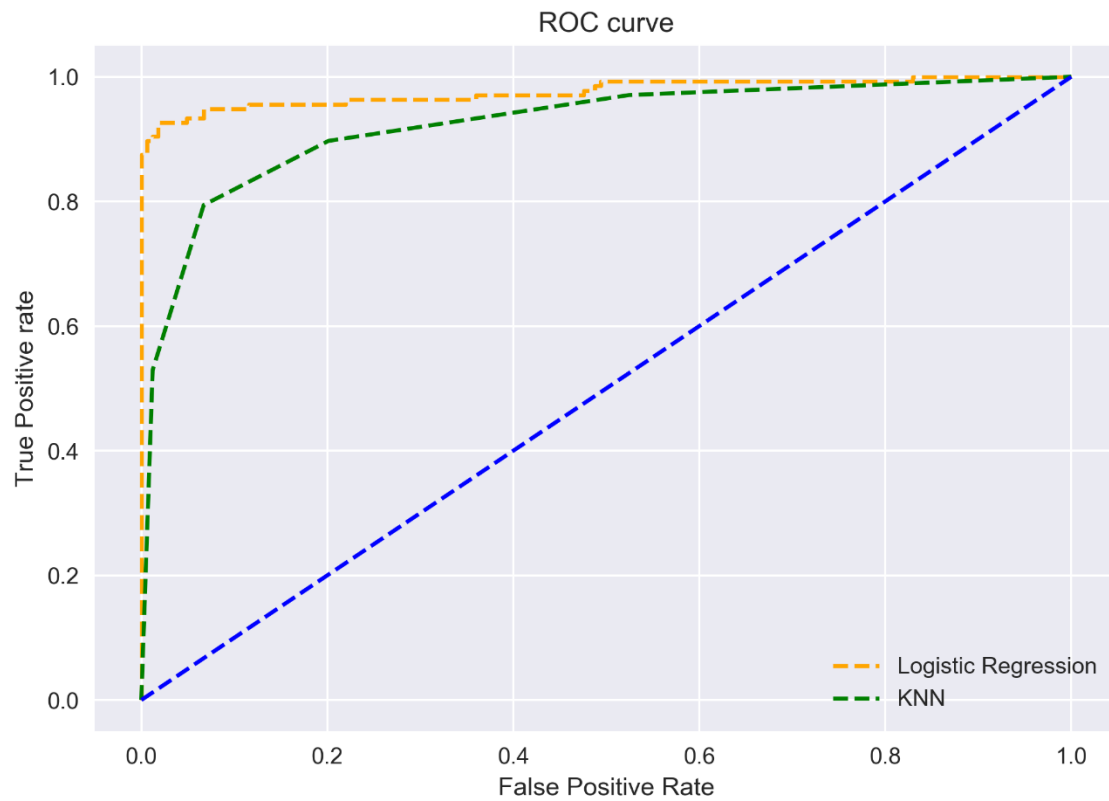
1.7 Using the confusion matrix, compute the following metrics: accuracy, true positive rate, false positive rate, and precision.

1.8 A model was trained on a training set for a binary classification task. The model's **probability** predictions on the test set is given in the table below. Compute the Precision-Recall points as you vary the threshold. Then plot the PR curve for this model, and be sure to plot the baseline precision.

test set

Feature 1	Feature 2	True Label	Model D Predicted Prob($y = 1 \mid x$)
1.09	0.62	1	0.77
1.18	0.48	0	0.74
1.18	-0.99	1	0.71
1.81	-1.4	0	0.71
2.48	-2.48	0	0.67
0.32	2.57	1	0.65
0.36	0.75	0	0.62
0.17	2.82	0	0.6
-0.23	-1.77	0	0.4
-0.32	-1.55	0	0.36
-0.35	-0.34	1	0.34
-0.41	1.25	0	0.3
-0.62	0.91	0	0.24
-0.78	0.49	0	0.21
-0.93	0.94	0	0.17
-1.7	1.19	0	0.07
-1.66	1.06	0	0.07
-2.08	1.26	0	0.04
-2.14	-0.48	0	0.04
-2.31	1.12	1	0.03

1.9 Consider the following ROC curve plots for two models (Logistic Regression and k-NN). Which model is better based on the ROC curves. Explain briefly why.



RL:

Recall the 1x4 example given in the slides with four states A, B, C, G. The agent starts in A.

Goal: G (terminal, reward 0). Rewards: -1 for every non-terminal step

Actions: Left, Right. 60% success, 40% stay in place.

Gamma: 0.9

A	B	C	G
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State	Action	$p(s', r s, a)$
A	Right	0.6 (B); 0.4 (A)
A	Left	0.6 (A); 0.4 (A)
B	Right	0.6 (C); 0.4 (B)
B	Left	0.6 (A); 0.4 (B)
C	Right	0.6 (G); 0.4 (C)
C	Left	0.6 (B); 0.4 (C)

This is the current V and π .

V_c

$V(A)$	-2.7100
$V(B)$	-2.4184
$V(C)$	-1.4896

π_c

$\pi(A)$	←
$\pi(B)$	←
$\pi(C)$	→

Using these values, Q1 to do policy iteration, and Q2 to do value iteration.

$$\begin{aligned}
v_{k+1}(s) &\doteq \mathbb{E}_{\pi}[R_{t+1} + \gamma v_k(S_{t+1}) \mid S_t = s] \\
&= \sum_a \pi(a|s) \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_k(s')] \\
\pi'(s) &\doteq \operatorname{argmax}_a \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_{\pi}(s')]
\end{aligned}$$

Q1. We want to continue with 1 more iteration of **policy iteration**. So, do 1 policy eval step and 1 policy improvement step.

Note, during the eval step, in practice you would do many sweeps until convergence, however, just make 1 sweep to the states A, B, C.

Using $\pi(A) = \text{Left}$:

$$0.6 * [-1 + 0.9 * V(A) = -2.7100] = -2.0634$$

$$0.4 * [-1 + 0.9 * V(B) = -2.7100] = -1.3756$$

$$\text{new } V(A) = -3.4390$$

Using $\pi(B) = \text{Left}$:

$$0.6 * [-1 + 0.9 * V(A) = -2.7100] = -2.0634 \quad (\text{note we use old } V \text{ values})$$

$$0.4 * [-1 + 0.9 * V(B) = -2.4184] = -1.2706$$

$$\text{new } V(B) = -3.3340$$

Using $\pi(C) = \text{Right}$:

$$0.6 * [-1 + 0.9 * V(G) = 0.0000] = -0.6000 \quad (\text{note that } V(\text{terminal state}) \text{ fix to } 0.0)$$

$$0.4 * [-1 + 0.9 * V(C) = -1.4896] = -0.9363$$

$$\text{new } V(C) = -1.5363$$

V_c after 1 eval sweep: {'A': -3.439, 'B': -3.334, 'C': -1.5363, 'G': 0.0}

Now do policy improvement.

State A:

Action 'Left':

$$0.6 * [-1 + 0.9 * V(A) = -3.4390] = -2.4571$$

$$0.4 * [-1 + 0.9 * V(A) = -3.4390] = -1.6380$$

$$\text{sum} = -4.0951$$

Action 'Right':

$$0.6 * [-1 + 0.9 * V(B) = -3.3340] = -2.4004$$

$$0.4 * [-1 + 0.9 * V(A) = -3.4390] = -1.6380$$

$$\text{sum} = -4.0384$$

argmax: Right

State B:

Action 'Left':

$$0.6 * [-1 + 0.9 * V(A) = -3.4390] = -2.4571$$

$$0.4 * [-1 + 0.9 * V(B) = -3.3340] = -1.6002$$

$$\text{sum} = -4.0573$$

Action 'Right':

$$0.6 * [-1 + 0.9 * V(C) = -1.5363] = -1.4296$$

$$0.4 * [-1 + 0.9 * V(B) = -3.3340] = -1.6002$$

$$\text{sum} = -3.0298$$

argmax: Right

State C:

Action 'Left':

$$0.6 * [-1 + 0.9 * V(B) = -3.3340] = -2.4004$$

$$0.4 * [-1 + 0.9 * V(C) = -1.5363] = -0.9531$$

$$\text{sum} = -3.3534$$

Action 'Right':

$$0.6 * [-1 + 0.9 * V(G) = 0.0000] = -0.6000$$

$$0.4 * [-1 + 0.9 * V(C) = -1.5363] = -0.9531$$

$$\text{sum} = -1.5531$$

argmax: Right

New policy: {'A': 'Right', 'B': 'Right', 'C': 'Right'}

$$\begin{aligned}
v_{k+1}(s) &\doteq \max_a \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) \mid S_t = s, A_t = a] \\
&= \max_a \sum_{s', r} p(s', r \mid s, a) \left[r + \gamma v_k(s') \right],
\end{aligned}$$

Q2. We want to continue with 1 more iteration of **value iteration**.

In practice you would do many sweeps until convergence, however, just make 1 sweep to the states A, B, C.

State A:

Action 'Left':

$$0.6 * [-1 + 0.9 * V(A) = -2.7100] = -2.0634$$

$$0.4 * [-1 + 0.9 * V(A) = -2.7100] = -1.3756$$

$$\text{sum} = -3.4390$$

Action 'Right':

$$0.6 * [-1 + 0.9 * V(B) = -2.4184] = -1.9059$$

$$0.4 * [-1 + 0.9 * V(A) = -2.7100] = -1.3756$$

$$\text{sum} = -3.2815$$

$$\text{max over actions: Right} \rightarrow V(A) = -3.2815$$

State B:

Action 'Left':

$$0.6 * [-1 + 0.9 * V(A) = -2.7100] = -2.0634$$

$$0.4 * [-1 + 0.9 * V(B) = -2.4184] = -1.2706$$

$$\text{sum} = -3.3340$$

Action 'Right':

$$0.6 * [-1 + 0.9 * V(C) = -1.4896] = -1.4044$$

$$0.4 * [-1 + 0.9 * V(B) = -2.4184] = -1.2706$$

$$\text{sum} = -2.6750$$

$$\text{max over actions: Right} \rightarrow V(B) = -2.6750$$

State C:

Action 'Left':

$$0.6 * [-1 + 0.9 * V(B) = -2.4184] = -1.9059$$

$$0.4 * [-1 + 0.9 * V(C) = -1.4896] = -0.9363$$

$$\text{sum} = -2.8422$$

Action 'Right':

$$0.6 * [-1 + 0.9 * V(G) = 0.0000] = -0.6000$$

$$0.4 * [-1 + 0.9 * V(C) = -1.4896] = -0.9363$$

$$\text{sum} = -1.5363$$

max over actions: Right $\rightarrow V(C) = -1.5363$

V updated to: {'A': -3.2815, 'B': -2.675, 'C': -1.5363, 'G': 0.0}

Notice in value iteration we only maintain V; in policy iteration we maintain both V and π .

Once we are done value iteration, how to get the final policy from final V? Just use the pi formula below.

$$\pi'(s) \doteq \operatorname{argmax}_a \sum_{s', r} p(s', r | s, a) \left[r + \gamma v_{\pi}(s') \right]$$

So from the current V, the policy would be: {'A': 'Right', 'B': 'Right', 'C': 'Right'}

Notice that both policy iteration and value iteration produce deterministic policies.